A Modeling Approach for Time-Dependent Geometry Applied to Transient Heat Transfer of Aluminum Electrolysis Cells

FRANÇOIS ALLARD, MARTIN DÉSILETS, and ALEXANDRE BLAIS

The thermal balance of aluminum electrolysis cells (AEC) have to be rigorously controlled in order to improve the efficiency and sustainability of this industrial process. A new modeling strategy is developed to consider the displacements of solid bodies and moving boundaries in finite element models. The transient thermal-electric modeling of the AEC demonstrates the effect of an increase in operating voltage on both the anode cover and the side ledge. With higher heat generation, the anode cover deteriorates and the side ledge thickness decreases. Since the anode cover is characterized by irreversible transformations, the top heat dissipation remains higher even when the operating voltage comes back to its typical value. For the first time, the transient temperature and electric fields throughout the anode life are simulated and validated by industrial measurements. The modeling predictions have been validated from instrumented anodes and manual measurements, all performed on operating AEC.

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I. INTRODUCTION

THE aluminum electrolysis process is continuously improving in order to reach a better energetic efficiency and environmental balance, while increasing the production at the same time. To achieve these goals, the design of the aluminum electrolysis cells (AEC) is optimized by performing modeling and experimental investigations. The industrial aluminum production is carried out by reducing alumina (Al_2O_3) with carbon anodes in a liquid electrolytic bath. The liquid bath is primarily composed of a mixture of Na_3AlF_6 (cryolite), AlF_3 (aluminum fluoride), CaF_2 (calcium fluoride), and dissolved Al_2O_3, but it usually also contains additives in the form of lithium, potassium, and magnesium fluoride salts. Eq. [1] is the overall electrochemical reaction occurring in AEC[1]:

$$2Al_2O_{3(\text{dissolved})} + 3C_{(s)} \overset{\text{current}}{\longrightarrow} 4Al_{(l)} + 3CO_{2(g)}. \quad [1]$$

The liquid aluminum is continuously produced by the electrochemical process and is extracted periodically by siphoning during industrial operations. Carbon dioxide (CO_2) is produced by the reaction of oxide ions at the surface of the carbon anodes immersed in the liquid bath. In order to feed the electrochemical reaction, an electric current is supplied to the AEC and is divided between the several consumable carbon anodes. The total current intensity varies from 200 to 600 kA depending on the technology. The main components of the AEC are described in Figure 1.

The heat balance of the AEC is rigorously controlled in order to maintain a protective layer of frozen bath, usually called side ledge, on the side walls. This layer protects the side components from the highly corrosive and erosive behavior of the liquid bath. The thermal conductivity of the side ledge has been recently determined by laser flash analysis at temperature up to 550 °C.[2–4] The side ledge grows by the primary crystallization of Na_3AlF_6 at a temperature close to 950 °C, while the AEC operates with a bulk liquid bath at around 965 °C. The side ledge formed at equilibrium conditions demonstrates a composition close to pure Na_3AlF_6, while with higher cooling rate the AlF_3, CaF_2, and Al_2O_3 species from the bath can be trapped inside its microstructure.[5–7] A ledge toe is also formed on the cathode blocks by both the solidification of Na_3AlF_6 and the precipitation of Al_2O_3 at temperature also close to 950 °C, depending on the local chemical composition.[6] When the AEC is quickly cooled down, the side ledge grows by forming an open crystalline layer, in which mass transfer can occur with the bulk liquid

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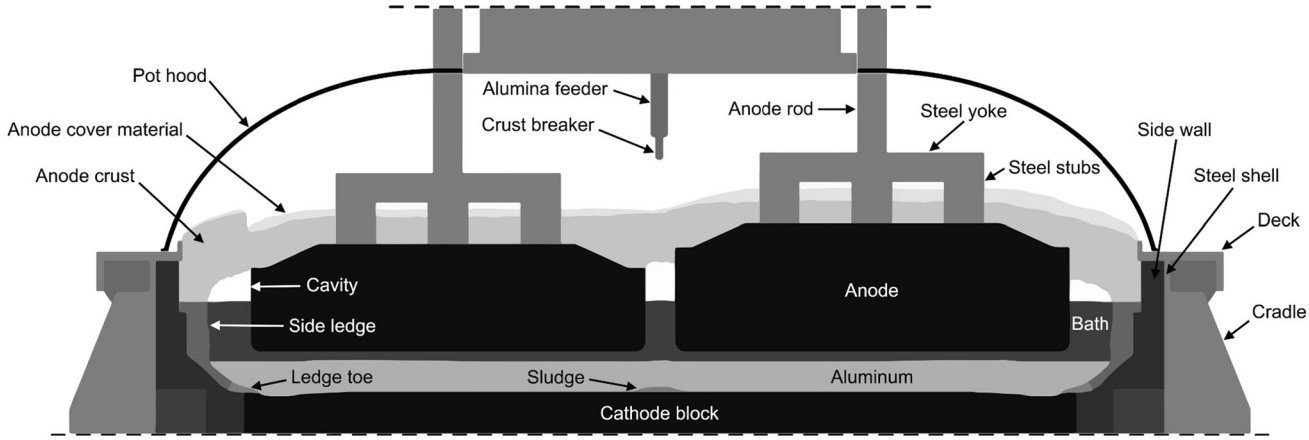


Fig. 1—An aluminum electrolysis cell with its main components.

bath.[8] The side ledge and ledge toe are modeled with their typical composition, thermal property, and behavior, which is expected to be close to chemical equilibrium.

Since the carbon anodes are consumed by the electrolysis process, they must be replaced after a few weeks of operations. The anode change procedure consists of breaking the anode crust, removing the old anode and installing a new one. During the first hours of the newly set anode, the top part of the liquid bath surrounding this anode is freezing in contact with colder gas. Once a "top freeze" layer of a few centimeters is formed, the anode cover material (ACM) is poured on the carbon anode and in the channels around the new anode. The ACM is leveled to prevent the oxidation of the carbon anode and to insulate the top of the AEC. Subsequently, the temperature increases at the bottom of the anode cover due to its insulating property, up to a point where the top freeze layer melts. At this step, the liquid bath from the electrolysis cell begins to rise in the powdered ACM by capillary forces. The ACM is transformed into anode crust when in contact with the liquid bath and vapor coming from the cell, at a temperature range between 685 °C to 710 °C depending on the local conditions.[9–11] Once formed, the hard anode crust is composed of α-Al_2O_3 particles cemented together with solid or liquid bath.[12–14] The anode crust should remain in place during all the life of the carbon anode, but its structure deteriorates due to liquid bath splashes and vapor penetration from beneath. At a temperature close to 935 °C, all the bath phases composing the anode crust are liquid and only the network of Al_2O_3 particles supports this material. Consequently, a cavity filled with fumes and gases will be formed when this part melts and collapses.[15]

The anode cover of AEC is defined by two distinct parts: the ACM located above and the anode crust underneath. The ACM typically consists of alumina, electrolytic bath, and recycled anode crust which have been crushed into particles of fine to coarse granulometry. The thermal properties of the ACM and anode crust have been investigated by several researchers.[9,16–21] The thermal conductivity of the ACM increases with both the temperature and the bulk density.[16,17,21] The composition of the ACM affects the thermal conductivity of the anode crust, since a higher proportion of crushed bath in the initial mixture forms an anode crust of higher thermal conductivity.[18] Unlike the ACM, anode crusts sampled in industrial cells demonstrate a slight decrease of thermal conductivity with increasing temperature.[19] Recent work reported the thermal conductivity of industrial anode crust measured by the laser flash method, which decreases from 1.38 to 0.89 W/m °C at temperatures ranging from 100 °C to 950 °C.[9] The emissivity of the ACM and anode crust has been evaluated experimentally and showed a value close to a black body.[10] The recent efforts to improve the thermal properties and geometries of the ACM and anode crust enable better predictions obtained with heat transfer models in steady state.[10]

The design and operation of AEC have been investigated by various numerical models taking into account different phenomena. Modeling of the mechanical stresses in the cell components, their heat generation and dissipation, the magnetohydrodynamic flow, the voltage drops in conductors, and the mass transfer in the liquid electrolyte have been investigated by many researchers.[22–30] However, most models that relate to thermal-electric modeling are in steady state and use a simplified representation of the anode cover. The thermal-electric modeling in transient state was applied to both preheating and cooling of stopped AEC.[27,31,32] More recently, a 2D finite difference model of the anode cover was built to investigate the transient heat transfer and the effect of the liquid bath temperature on the first days of the crust formation.[33] The impact of the anode change on the temperature profile and fluid flow, considering the first day of anode life, was investigated with computational fluid dynamics (CFD) simulations.[34] A frozen bath layer is formed on the cold anode when it is inserted into the liquid bath. Its crystalline microstructure is rather heterogenous and presents large pores caused by entrapped gas and shrinkage of the sample, according to scanning electron microscopy and tomography analyzes.[35]

The top parts of the anode assembly and cover are cooled by air from the potroom (a large building housing the AEC) that enters through the gaps between the pot hoods due to the negative pressure created by the pot ventilation system. The top heat losses increase when the pot ventilation rate is raised, while the gas temperature leaving by the duct will reduce.[36] The convection coefficients applied to the top parts depend on the volumetric flow rate of gas collected from the pot according to CFD simulations.[37] The convection heat transfer can be varied by adjusting the gas flow rate, but a sufficient pot tightness must be kept to avoid leakage of harmful gases to the environment.[38] In the computation of flow or fluid–structure interaction, automatic mesh update strategies were previously developed and integrated for these numerical problems.[39,40] The moving mesh or automatic remeshing allows to resolve problems with moving parts, boundaries, and interfaces. However, these techniques are still not available for the thermal and electric models solved with the finite element method, to the authors' knowledge. A modeling approach for moving boundaries and time-dependent geometry is developed to resolve the heat transfer and electric conduction taking place in the industrial electrolysis cell.

The goal of thermal-electric modeling is to optimize the design and operation of AEC in order to reach higher efficiency. While the steady state analysis is useful to predict the overall behavior of an AEC, the transient analysis enables the investigation of time-dependent phenomena. In this study, two transient thermal-electric models are developed to (1) predict the effect of an electric potential (voltage) increase on the heat balance and (2) evaluate the behavior of an anode throughout its life cycle. The first numerical model quantifies the effect of a voltage increase on both reversible and irreversible phase change materials, which control the thermal behavior of the electrolysis cell. Moreover, industrial measurements of electric potentials, material thicknesses, and temperatures were performed in industrial AEC to validate the model's predictions. A second transient thermal-electric model is developed in order to predict the thermal dissipation and the dynamic of the anode cover during the anode life cycle. This new model uses a mathematical algorithm to account for the anode consumption and movement in time, while solving the heat and electric conduction. Since this domain is time-dependent, a strategy is developed for the moving geometry, mesh update, and nodal interpolation and is applied to the numerical model. Finally, the prediction results are thoroughly validated with continuous measurements from instrumented anodes installed in an industrial AEC.

II. TRANSIENT THERMAL-ELECTRIC MODELS

The transient thermal-electric models predicted the heat transfer and electric potentials of a P155 prebake cell. The simulations were validated by measurements in industrial cells operated by Rio Tinto's Grande-Baie plant. The measurements have been obtained during the fall season and no special events were noticed during the operation of the AEC. Two instrumented anodes were continuously monitored during their operation for the purpose of this study. One evolved with normal conditions, while the other experienced a voltage increase (defined as "voltage treatment" for this study).

The steady state modeling method and the thermal properties of the top parts have been recently enhanced in a recent work; these articles may be consulted for further details.[9,10] These improvements were also applied to the transient models presented here. The modeling methodology will be described with a peculiar emphasis on new modeling methods and thermal properties. The following two transient models, with a geometry and mesh updating algorithm, are developed and detailed in this study:

(1) Slice of AEC
(2) Anode life cycle

A. *Geometry of the Slice of AEC*

The domain for the first model is a slice of an industrial AEC containing one and a half anodes and two cathode blocks. The heat transfer and the electric field are predicted inside the following boundaries: from the top of the anode cover, including anode assemblies, to the bottom of the AEC with its steel shell and cradles (shown in Figure 2). The top parts include the anode assembly (anode rod, yoke, stubs, and carbon anode) and the anode cover (ACM and anode crust). A cavity located between the anode crust and the bulk liquid bath is integrated in the geometry. The cavity extends all

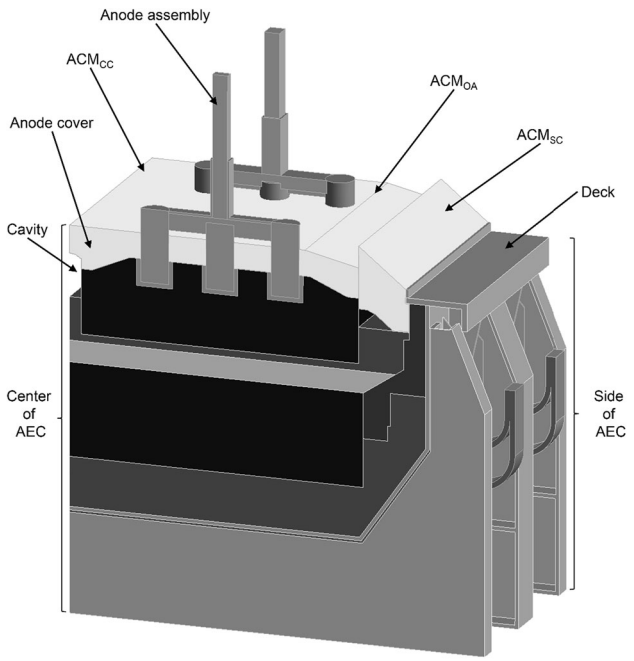


Fig. 2—The slice geometry for an aluminum electrolysis cell, as used for the first model.

around the anodes: in the side channel (SC), between the anodes. and in the center channel (CC). The superstructure, anode beam, and hoods are excluded from the slice geometry which comprises most parts of an industrial AEC, starting from the anode rod to the flexible conductors on the collector bar. For this model, mid-life anodes are used to replicate the overall behavior of the industrial AEC.

The ACM and anode crust profiles on mid-life anodes were measured in industrial cells with an experimental device using the method shown in Figure 3(a). A steel bar was positioned at each location required to measure the surfaces of ACM and anode crust. The deck was the reference point to locate the absolute height of these materials. The height of the cavity was measured by drilling a hole at the location (B) in order to validate the model prediction. The crust–cavity interface (C) is predicted by the transient model, at each analysis step, to take into account the anode crust deterioration. This boundary moves according to the heat balance and the thermal excursions occurring in AEC, which increase the local temperature and cause the melting of the anode crust.

The anode crust in industrial cells can typically be divided into two distinct parts, the breaking point being located at the extremity of the carbon anodes, toward the side of the pot. The anode beam movement generates mechanical stresses on the anode crust resulting in the formation of the crust_{SC} (see Figure 2), which remains attached to the side wall throughout the anode life. The shape of crust_{SC} is similar to a "step" after a few days of the anode life. The crust on the anode (OA) and crust_{CC} are linked to the anode and follow its movement.

The top geometry of the anode cover is set based on previous measurements of mid-life anodes in industrial AEC.[10] The angles for the top surfaces of the anode cover are given in Table I. The distribution of both ACM and anode crust in the anode cover is calculated by the transient model and it depends on the corresponding transformation temperature range. Thus, the profiles of ACM and anode crust are predicted by the transient model.

The bottom surface of the anode crust is complex to measure due to the local conditions: high temperature, harmful gas, and corrosive bath splashes. Accordingly, the crust–cavity profile is predicted with transient modeling. This profile is time-dependent when the steady state conditions no longer apply, such as during voltage change or anode consumption. A new geometry of crust is calculated after each analysis step to consider the melting of the anode crust. The new geometric parameters of the crust bottom surface are modified automatically in the computer-aided design (CAD) tool to feed the next step.

B. *Modeling Method for the Slice of AEC*

The thermal balance of AEC mainly relies on the heat generation due to Joule heating effect and its dissipation to the environment. The electric current enters from the top conductors and leaves through the collector bars, but the heat generation is mainly produced in the resistive liquid bath located between the anode and the cathode. The transient heat transfer is predicted with the coupled thermal-electric model applied to the slice of AEC (Figure 2). This model is resolved with the finite element method and the domain is discretized into approximately 700,000 elements with midside nodes

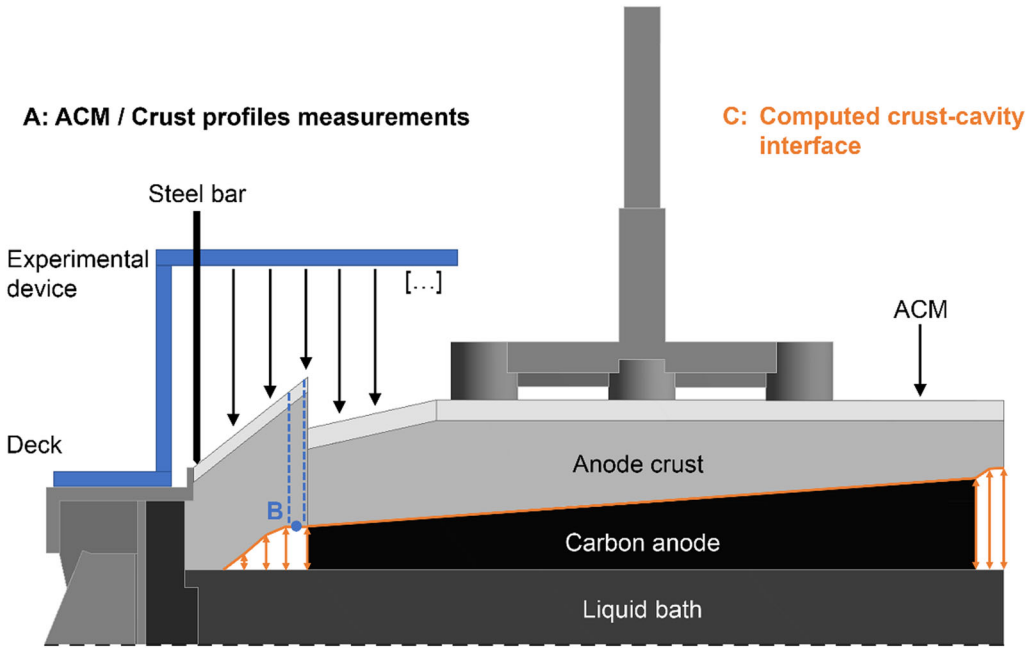


Fig. 3—(*a*) ACM and anode crust profiles measurements methodology in industrial cells. (*b*) Blue-filled circle indicates the location where the height of the cavity was measured. (*c*) The crust–cavity interface facing the bulk liquid bath which is calculated for each model (Color figure online).

Table I. Geometric Parameters of the Anode Cover of Mid-Life Anodes

Angle (°)*	
ACM_{SC}	ACM_{OA}
35	15

*Angle from the horizontal.

(tetrahedrons). The commercial software ANSYS Mechanical (release 18.0) was used to perform the thermal-electric calculations using an Intel Xeon E5-1650 v4 with 32 GB ECC memory.

A cavity is added in the model to consider the geometric aspects of the anode cover and to improve the heat transfer prediction. Figure 3(c) demonstrates the crust–cavity boundary, where radiation heat transfer is calculated. The cavity is integrated with a radiosity module that calculates the outgoing radiative flux for each surface (carbon anodes, liquid bath, and bottom of the anode crust). This module is solved in conjunction with the thermal-electric problem. The crust–cavity interface is displaced based on the melting temperature of the chemical compounds in the anode crust. Accordingly, a moving boundary located at the bottom of the anode crust is displaced to reach the melting isotherm in the domain according to the predicted nodal temperatures. Then, the prediction of the crust–cavity profile allows to quantify the impact of operating parameters on the heat loss distribution.

The temperatures and electric fields are predicted by a step-by-step method which is automated using Python 2.7 scripting language in ANSYS Workbench. The programmed method updates the geometry and the mesh, and then applies the temperatures from the previous step to the new mesh. The detailed explanations of the modeling procedure can be found in "Appendix A" (see Figure A1). The model can be initialized from a previous steady state prediction by importing the nodal temperatures or by performing a new transient analysis. In summary, the model calculates the nodal temperatures and electric potentials for a given analysis time and export these results to the next geometry with a new crust–cavity profile. The exported nodal results contain the locations (x, y, z) and the temperatures (T) and these values are applied to the new generated mesh as initial conditions mapped with a 3D interpolation method. This procedure is repeated up to the final time of the analysis. The irreversible ACM to anode crust transformation is also integrated in the modeling algorithm by saving the previous material states of the elements in a file and by applying the values to the newly generated elements. The geometry and mesh update method described in this study is not subject to the problem of mesh distortion. However, this problem can be expected to happen with a moving mesh method, as described in Reference 41.

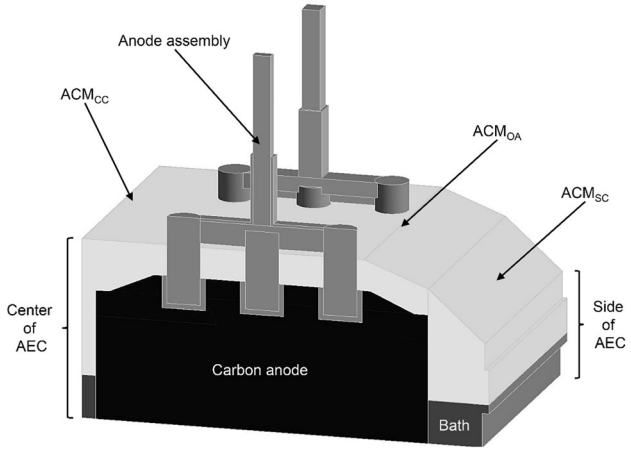


Fig. 4—The geometry of the anode life cycle model (top parts of the AEC).

C. *Geometry for the Anode Life Cycle Modeling*

The anode life cycle is modeled using the top parts of the AEC (see Figure 4) in order to investigate the thermal dissipation and the evolution of the anode cover. An industrial AEC contains many anodes and its thermal behavior depends on the overall top heat losses, and thus the bath and ledge temperatures need to be first predicted using the previous model of the slice of AEC. A steady state analysis has been performed to get the temperature profile in the bulk liquid bath and the side ledge. These results were applied by nodal constraints in the bath of the anode life cycle model. The profiles of ACM and anode crust are also predicted with this transient model.

The top boundaries and materials are similar to the first model, but the main geometrical differences are the carbon anode consumption and the anode assembly displacement. The ACM and anode crust located OA and in the CC are attached to the anode and follow its motion.

Once transformed, the anode crust deteriorates and a cavity starts to grow. Accordingly, a radiosity module is also integrated in this model and the anode crust–cavity boundary is moved according to the melting isotherm. The ACM and anode crust profiles were measured with the methodology previously described in Figure 3(a) in order to set the initial ACM geometry and to ensure that the transient model represents the real evolution correctly.

D. *Modeling Method for the Anode Life Cycle*

A modeling approach is developed in order to predict the thermal-electric field for all the life cycle of the carbon anode. In the industrial process, a beam supporting all the anode assemblies is lowered to follow the carbon consumption and the operations (per example, siphoning). The parts of the anode crust, located OA and in the CC, track the anode displacement, while the crust on the SC region remains attached to the side wall. Additionally, the bottom of the anode crust melts to

form the cavity. This second model aims at predicting the thermal dissipation from the geometrical variations of the anode cover and carbon content during its life cycle. The domain is discretized into approximately 600,000 elements with midside nodes (tetrahedrons).

Since the second model has a time-dependent domain and a moving boundary, a method is established to simulate these conditions in conjunction with the transient thermal-electric problem. A strategy for moving geometry, mesh update, and nodal interpolation is developed and integrated in the ANSYS Mechanical software. This approach is detailed in "Appendix B" (see Figure B1). The time-dependent domain is accounted by exporting the nodal properties to a matrix and by calculating the new coordinates depending on the chosen analysis time, since the velocity of geometric displacement is known. Subsequently, the mesh is updated and the newly calculated nodal properties stored in the matrix are applied as initial conditions to the new mesh with a 3D interpolation method. In parallel, the moving boundary is treated by the same mean as described in Section II–B. This step-by-step approach is programmed to automate the preprocessing, resolution, and postprocessing in the ANSYS commercial software. Although this approach is currently applied to thermal-electric modeling, it can also be convenient to describe other similar physical phenomena, or to improve the capacities of other commercial software. Moreover, the moving geometry, mesh update, and nodal interpolation approach does not exhibit mesh deformation due to the mesh update step. Effectively, a simple moving mesh technique would result in important mesh distortion, considering these specific domain changes, and is therefore not advisable in this work.

In order to demonstrate this modeling approach, only the top parts of the AEC have been selected in the CAD geometry. Since the anode size and its movement are time-dependent, they are modeled using the moving geometry, mesh update, and nodal interpolation strategies. The irreversible phase transformation of the anode crust is also integrated in the modeling algorithm. For the moving parts, the nodal and element coordinates are displaced using matrix operations. In this simulation case, the bottom frontiers are the liquid bath and the side ledge (see Figure 4). Their nodal temperatures have been previously calculated with the slice model and they were applied as constraints for the anode life cycle modeling.

E. *Assumptions for the Transient Modeling*

In order to compute the transient thermal-electric solutions of both the slice of AEC and the anode life cycle, the following assumptions are considered: (1) ACM and anode crust behaviors are controlled by heat transfer and thermochemical processes; (2) the temperature and enthalpy of ACM → anode crust → cavity transformations are similar to already published results[9]; (3) the phase transitions inside these materials evolve to reach a state of chemical equilibrium, and thus the reactions can be predicted with Gibbs energy minimization; (4) the heat transfer associated to mass transport is neglected; (5) the ACM granulometry, density, and thermal conductivity are similar to the data presented in Reference 17; (6) the thermophysical properties of all materials are known; (7) the effect of the flow circulation inside the bulk liquid bath and the aluminum pad is accounted by an "augmented effective thermal conductivity"[42]; (8) the heat transfer in the cavity is radiation dominated; (9) the surfaces in the cavity are gray and diffuse; and (10) the time-variation of the electric field due to permittivity is neglected.

F. *Governing Equations*

The transient heat transfer and heat generation are predicted based on a coupled thermal-electric analysis. A global system of algebraic equations is obtained with an assembly of finite elements. In this case, the finite element method solves the differential and integral equations using 3D coupled-field elements for solid and surface elements for boundary conditions. The set of algebraic equations is developed with the weighted residual statement and the Newton–Raphson method is applied to solve this nonlinear system. The preconditioned conjugate gradient (PCG) solver assembles the full stiffness matrix and it calculates the solution by iterating up to convergence. The ANSYS Mechanical solver available with APDL (ANSYS Parametric Design Language) script is used to resolve each simulation case.

Equation 2 describes the governing equation for the transient heat conduction with the Joule heat generation rate per unit volume ($\dot{q}$)[41,43]:

$$\rho c_p \left(\frac{\partial T}{\partial t}\right) + \frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_z \frac{\partial T}{\partial z}\right) = \dot{q}. \quad [2]$$

For a transient or steady state electric analysis, the Joule heat in a conductor is described by Eq. [3] using the centroidal values of the electric field and current density:

$$\dot{q} = \boldsymbol{J} \cdot \boldsymbol{E}. \quad [3]$$

The electric field can be derived from

$$\boldsymbol{E} = -\nabla V. \quad [4]$$

The electric potential distribution is computed from the governing equation for electric conduction, in which the electrical conductivity (σ) depends on the temperature of the solid:

$$-\nabla \cdot ([\sigma] \nabla V) = 0. \quad [5]$$

Further information on the thermal-electric coupling is available in the ANSYS Mechanical documentation.[43] The radiosity method is applied in order to calculate the heat transfer in the cavity formed between the bottom of the crust and the top of liquid bath. A set of linear algebraic equations is used

to obtain the outgoing radiative fluxes (radiosity) for each surface.

$$[A]\{q''^{,0}\} = \{D\}, \quad [6]$$

where

$$A_{ij} = \delta_{ij} - (1 - \varepsilon_i)F_{ij}$$
$$q_j''^{,0} = \text{radiosity flux for surface } j.$$
$$D_i = \varepsilon_i \sigma T_i^4$$

Eq. [6] is solved with the Newton–Raphson procedure for the radiosity flux $\{q''^{,0}\}$. At the surfaces facing the cavity, the boundary conditions for the heat conduction are provided by the net surface radiosity fluxes. The radiation view factors, F_{ij}, are required to solve the radiosity module. The overall view factors for each surface are calculated using the hemicube method.[43,44] Each surface on the hemicube has an overall view factor given by

$$F_{ij} = \frac{1}{A_i}\int_{A_i}\int_{A_j}\frac{\cos\varphi_i \cos\varphi_j}{\pi r^2}\mathrm{d}A_j\mathrm{d}A_i. \quad [7]$$

The result is a complete set of form factors:

$$F_{ij} = \sum_{n=1}^{N}\Delta F_n, \quad [8]$$

where N is the number of pixels, ΔF is the delta-view factor for each pixel.

G. *Material Properties*

The thermophysical properties have recently been improved and implemented in the transient thermal-electric models. Table II describes the properties with a focus on the new data and the top components of the AEC. The radiation heat transfer inside the cavity is predicted with a radiosity module considering the emissivity of the anode crust, the carbon anode, and the liquid bath. The bulk liquid bath and the aluminum pad are modeled with an augmented effective thermal conductivity to consider the flow circulation generated by the gas bubbles and the magnetohydrodynamics (MHD). A high ratio of advective to diffusive heat transport can be computed with the help of the Péclet number (Pe). Accordingly, the $k_{\text{liquid bath}}$ and the $k_{\text{liquid aluminum}}$ were both estimated to 10 000 W/m K to reflect the advective heat transfer due to the movements of these liquids.

The ρ_{ACM} was simply determined in the laboratory considering the mass and the bulk volume, which were measured using a scale and a graduated cylinder. The specific heat capacity (c_p) was determined from previous quantification data of a typical ACM and the average of some anode crusts.[9] The cryolite ratio (CR), which is the molar NaF/AlF_3 ratio, was 1.96 for the ACM and 1.77 for the anode crust. The $[Al_2O_3]$ for the ACM and the anode crust was 60 and 38 pct, respectively. Using their corresponding compositions, the theoretical c_p of these materials was calculated by chemical equilibrium calculations by means of the FactSage 7.1 software. Furthermore, the theoretical c_p has also been validated in a previous work with differential scanning calorimetry applied on industrial samples.[9] Figure 5 shows the temperature dependent c_p for the ACM and the anode crust, which also considers the enthalpy of phase transitions.

The ACM, the anode crust, and the side ledge are modeled as phase change materials. The ACM transforms into anode crust from 685 °C (Eq. [9]) due to the contact with liquid or vapor fluoride species coming from the bulk liquid bath. This transformation is also favored when the liquid fraction in this material increases. Accordingly, it enhances the growth of the α-Al_2O_3 crystalline network and consolidates the particles. The liquid fraction rises quickly from 696 °C to 710 °C in ACM of typical chemical composition, according to thermodynamic equilibrium results.

$$\text{ACM} \xrightarrow{\text{fluorides}} \text{anode crust} \quad (685\,^\circ\text{C to } 710\,^\circ\text{C}). \quad [9]$$

For the present simulations, the ACM thermal conductivity from room temperature up to 685 °C is represented by the linear equation determined by Shen.[17] A linear transition is included in order to reach the value of the anode crust (between 685 °C and

Table II. Thermophysical Properties of the Top Materials

Property	Value	Reference
ε_{ACM}	1.0	10
$\varepsilon_{\text{crust}}$	1.0	10
$\varepsilon_{\text{carbon}}$	0.95	45
$\varepsilon_{\text{liquid bath}}$	0.97	46
k_{ACM} (W/m K)	$0.000616T + 0.155$	17
k_{crust} (W/m K)	$-0.000541T + 1.386$	9
$k_{\text{solid bath}}$ (W/m K)	1.2	—
$k_{\text{liquid bath}}$ (W/m K)	10,000	—
ρ_{ACM} (kg/m^3)	1600	laboratory measurement
ρ_{crust} (kg/m^3)	2500	9
$c_{p,\text{ACM}}$ (J/kg K)	Fig. 5	chemical thermodynamic
$c_{p,\text{crust}}$ (J/kg K)	Fig. 5	chemical thermodynamic

T temperature in °C.

710 °C). From 710 °C and beyond, the crust irreversible transformation is considered completed and the thermal conductivity decreases almost linearly with temperature.[9]

Once the anode crust is formed, it begins to deteriorate due to high temperature and liquid bath splashes. From 935 °C, all bath phases are melted and the structure is only supported by the Al_2O_3 particles that are sintered together. The liquid bath splashes, which could have frozen at lower temperature, deteriorate the anode crust by dissolving Al_2O_3. Consequently, the anode crust will deteriorate with these conditions and fall down in the bulk bath, thus forming the cavity. The cavity is an empty space filled with process gases in which the radiation heat transfer takes place. Eq. [10] gives the irreversible anode crust to cavity transformation.

$$\text{Anode crust} \rightarrow \text{cavity} \quad (935\,^{\circ}\text{C}). \qquad [10]$$

The ledge is formed on the side walls when liquid bath freezes due to the heat dissipated by the shell to the surroundings. The ledge is primarily composed of Na_3AlF_6 with dissolved species inside its microstructure. The melting or freezing temperature of Na_3AlF_6 depends on chemistry of the bulk liquid bath, a phenomenon that affects the local composition at the liquid bath–ledge interface. The ledge grows or melts typically at 950 °C (Eq. [11]), a temperature that varies depending on the liquid bath composition. This reversible phase change evolves according to operating conditions and heat dissipation to the surroundings.

$$\text{Liquid bath} \leftrightarrow \text{side ledge} \quad (950\,^{\circ}\text{C}). \qquad [11]$$

The irreversible transformations in the anode cover, ACM → anode crust → cavity, are considered in the transient model by exporting the material conditions after each step of simulation. The values are stored in a CSV file and read in the next step. Afterward, a material property for each element is attributed based on the previous calculations. The irreversible anode crust to cavity transformation is integrated by means of the moving boundary method previously described in Section II–B.

H. *Boundary Conditions*

For the thermal-electric simulations, typical thermal boundary conditions are used for the side and the bottom of the slice of AEC, while the top thermal conditions are based on previously measured data.[10] The boundary conditions are established using the average value of heat flux sensors (HFS) and T_∞, which were measured previously throughout the follow-up of three instrumented industrial anodes. The top boundaries consist of the top surfaces of the ACM and anode assemblies. These parts are facing the bottom of the superstructure and the pot hoods which are at lower temperatures. Also, the air flow coming from the potroom and entering through the gaps between the hoods cools down the top parts by convection heat transfer. The top parts are then cooled by both convection and radiation heat transfer. An "effective convection heat transfer coefficient" (h_{eff}) considering both the convection and radiation has been applied to the top boundaries of the electrolysis cell. The h_{eff} data were calculated from the heat flux measurements (q'') with Eq. [12], knowing the surface (T_s) and gas (T_∞) temperatures.

$$q'' = h_{\text{eff}}(T_s - T_\infty). \qquad [12]$$

The boundary conditions, h_{eff} and T_∞, were determined based on thermal measurements in industrial cells (method detailed in Section II–A). T_∞ are the gas temperatures measured above the boundaries. Table III details the top boundary conditions used in the two numerical models.

The global geometry of the AEC was cut in the center and on two sides to produce the slice shown in Figure 2. For each cut section, a symmetric boundary condition has been applied. Heat losses from the side and the bottom were also predicted with h_{eff} and T_∞ data calculated and measured on industrial AEC.

For the slice of AEC, the current delivered to the electrolysis cell was divided by the number of anodes and the corresponding value was applied to the top of each anode rod. An electric potential of 0 V has been applied to the top of each cathode flexible conductor. For the anode life cycle model, the electric current was measured on an instrumented anode throughout its lifetime and the corresponding value was applied on the top of each anode rod.

I. *Measurements to Validate the Thermal-Electric Models*

The transient thermal-electric model of the slice of AEC is used to determine the nodal bath temperature and to apply these values as constraints in the anode life cycle model. Additionally, the effect of a voltage increase is investigated with the slice of AEC. In order to validate this heat transfer prediction, the instrumented anode (A) was monitored during the voltage

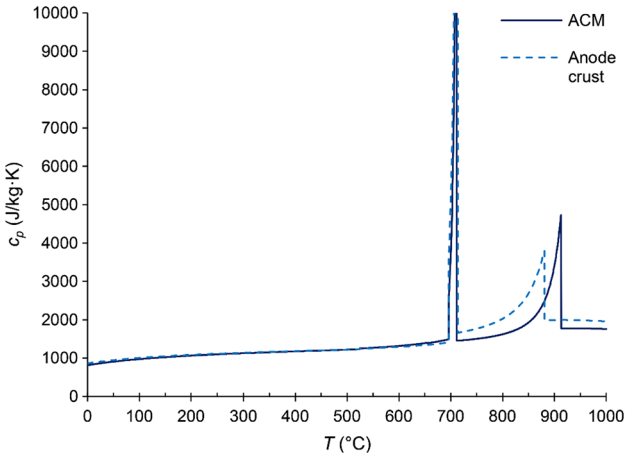


Fig. 5—Specific heat capacity (c_p) of the ACM and anode crust of typical chemical composition.

increase period. A type K thermocouple was installed on the side shell at the bath–metal interface (BMI) and another was kept in the liquid bath to evaluate the effect of the voltage increase on local temperatures. Moreover, the side ledge thicknesses were manually measured before and after this heat treatment. Another instrumented anode (B) was collecting data, while the AEC operated normally, thus without noticeable events. This second dataset was used to validate the anode life cycle model.

Figures 6 and 7 show the locations where the typical thermal boundary conditions have been previously determined with the manual measurements of heat flux and T_∞, which were performed throughout the life of the anodes.[10] The modeling results will be validated with the thermocouples installed on the instrumented anodes A and B investigated in this study (temperatures shown in Figure 7).

The heat fluxes and temperatures above the ACM and on some metallic surfaces have been determined from measurements with Hukseflux HF01 sensors (HFS). The estimated accuracy of these sensors is ± 15 pct and the repeatability is ± 5 pct according to the product manuals.[47] The temperature was also measured by each sensor with an accuracy of approximately ± 1 pct for their internally installed type K thermocouple.

The temperatures on the anode assembly, the ACM, and the anode crust were measured continuously with type K thermocouples (Omega) installed at the locations described in Figure 7. The thermocouples in the ACM and crust were installed before the covering of the new anode, folded in "L" shape, pointing in the opposite direction of the stubs. The ends of these thermocouples were located under the $HFS_{ACM,OA2}$, as shown in Figure 6. In the ACM-crust, a vertical arrangement of thermocouples was installed, with a sensor every 2.5 cm (from 0.0 to 10.0 cm) from the top of the carbon anode. On the anode rod, a thermocouple was soldered at each 10.0 cm. The gas temperatures (blue-filled circle) were measured manually with type *T* thermocouples at the locations shown in Figure 7. These gas temperatures were used to determine the boundary conditions previously presented in Table III.

Table III. Boundary Conditions for the Top Parts of the AEC

Boundary	h_{eff} (W/m^2 K)	T_∞ (°C)
Sides of anode rod	25	120
Steel yoke and stubs	40	120
Top of ACM_{OA} and ACM_{CC}	20	120
Top of ACM_{SC}	20	100

III. RESULTS AND DISCUSSION

A. *Effect of a Voltage Increase Investigated with the Slice of AEC*

The ACM → anode crust → cavity transformations throughout the anode life are irreversible, as explained in Section II–G. In industrial AEC, there are adverse conditions where the heat losses increase significantly for a period of time. When the AEC can be brought back to normal operation, the degradation of its anode cover subsists. In order to investigate the effect of a voltage increase and consequently a thermal excursion on heat losses, the operating voltage was increased while an instrumented anode was in the AEC to monitor the

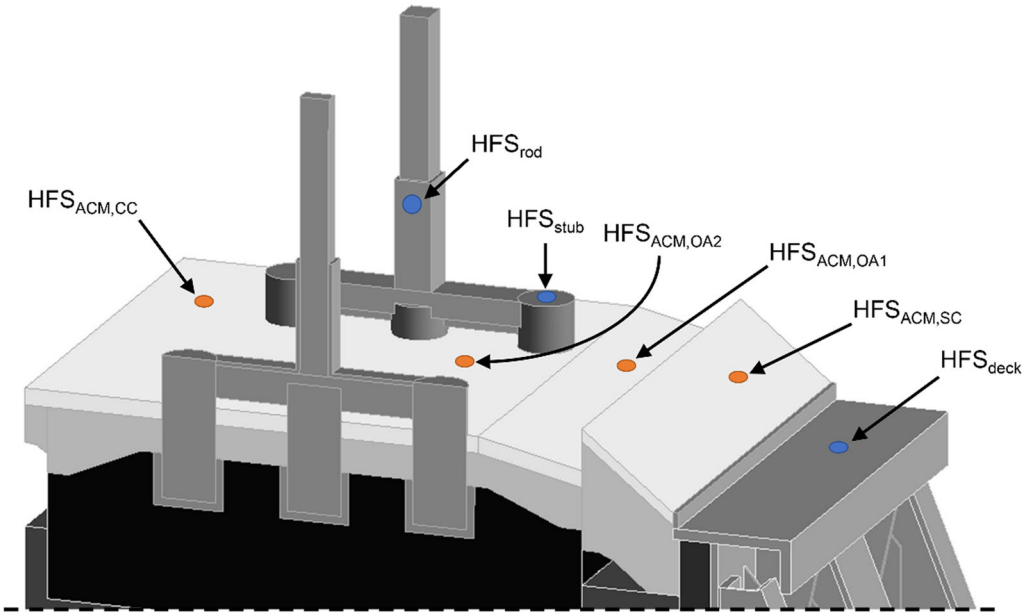


Fig. 6—Each (orange-filled circle) indicates the location on the ACM where a HFS was placed; blue-filled circle indicates the location of a HFS set on a metallic surface (Color figure online).

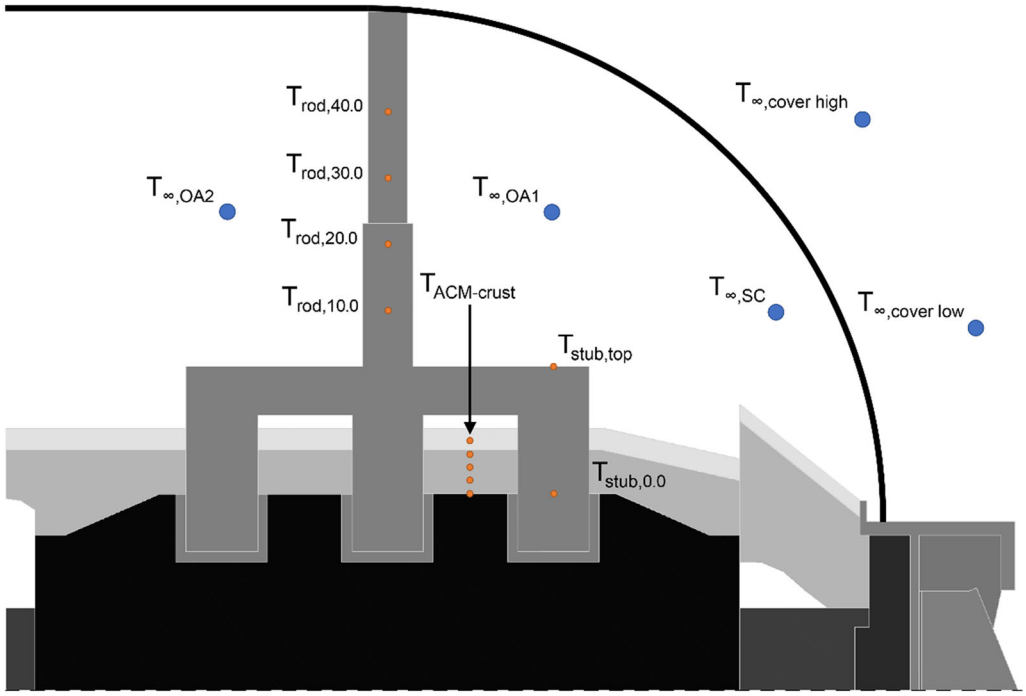


Fig. 7—Orange-filled circle: thermocouples installed in the ACM, anode crust, and anode assembly; blue-filled circle: locations where manual measurements of the gas temperatures were made (Color figure online).

changes taking place. A 8.5 pct voltage increase, calculated from the normal operating voltage, was applied for 23.0 h (0.958 day) on an industrial AEC, which was operating normally until this change. This voltage treatment was carried out by changing the operating set-point, which resulted in the increase of the anode-cathode distance (ACD). Among other things, this treatment was made in order to simulate an AEC experiencing a thermal excursion which generates the anode cover degradation and to evaluate the consequences of this event on the thermal balance.

The transient thermal-electric model of the slice of AEC was used to evaluate the effect of the voltage treatment and to determine the liquid bath and side ledge temperature profiles for the anode life cycle model (Section III–B). In order to predict the temperature and electric field during and after the voltage treatment, these cumulative steps were performed with the transient thermal-electric model:

(1) A steady state was predicted with the slice of AEC and used for the initial condition ($t = 0$).
(2) The voltage increase was evaluated up to $t = 0.958$ day.
(3) A normal voltage was applied again to assess the cell behavior from $t = 0.958$ day to $t = 3$ days.

The voltage increase was simulated by means of an electrical resistivity change (vertical component only) in the liquid bath located between the anode and the liquid aluminum. The transient modeling results were validated from the instrumented anode A and the temperature from the thermocouples on the side shell and in the liquid bath.

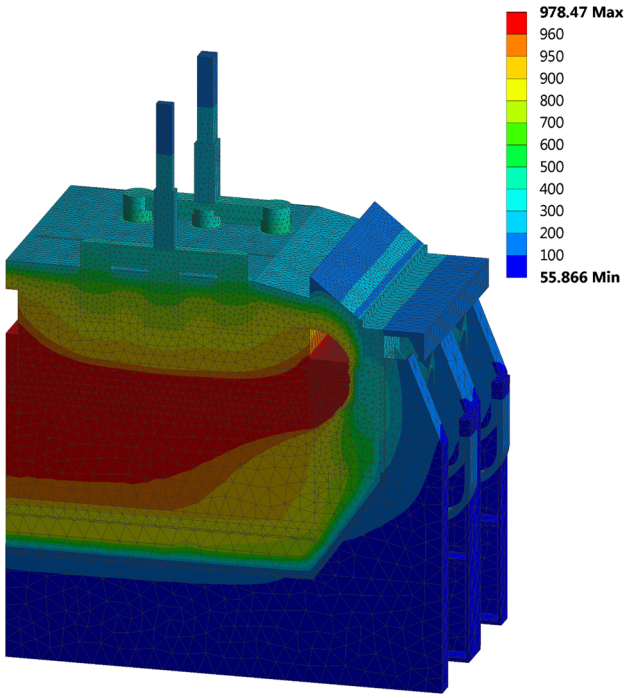


Fig. 8—Temperature profile (°C) predicted with the slice of AEC at $t = 0$ (initial condition).

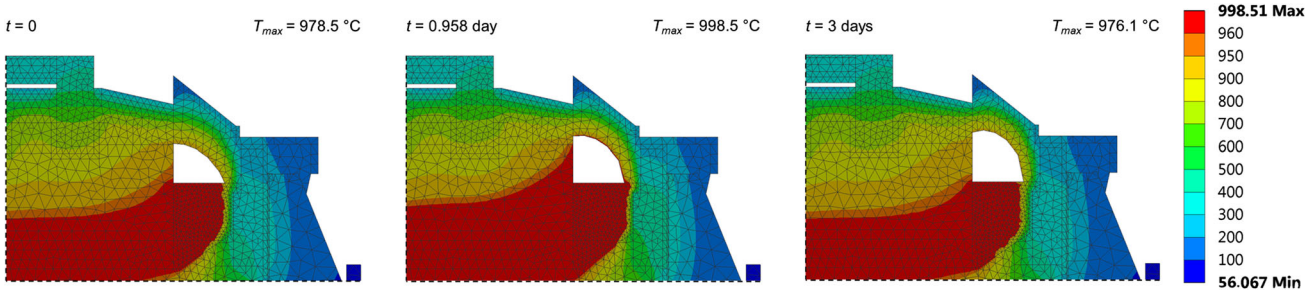


Fig. 9—Temperature profiles (°C) predicted with the slice of AEC at $t = 0$, $t = 0.958$ day and $t = 3$ days.

1. *Temperature profile*

The transient modeling with the slice of AEC began with a steady state ($t = 0$) in order to simulate an industrial AEC operating at stable conditions. This simulation considered the ACM to anode crust transformation by changing the material property of each element and also integrated the anode crust melting with the moving boundary. The corresponding temperature distribution is shown in Figure 8.

Then, the voltage increase (+ 8.5 pct) was applied for 23 h (0.958 day) and a stabilization period of 2.042 days was evaluated. The temperature profiles for these consecutive simulations are presented in Figure 9. At the end of the voltage treatment, T_{max} in the liquid bath reached 998.5 °C, which is 20 °C higher than the initial condition. Considering this high temperature, the radiation heat transfer from the liquid bath is enhanced, thus causing the melting of the anode crust. The cavity boundary moved up and its size increased considerably, as observed in Figure 9. After 3 days, T_{max} of the liquid bath decreased to 976.1 °C, the phase change materials have stabilized in time and a new steady state was reached.

2. *Effects on the side ledge, the cavity, and the ACM*

The effects of the voltage increase on the side ledge, the cavity size, and the ACM transformation are detailed in Table IV. The height of the cavity_{SC}, taken at a location shown in Figure 3(b), was 13.7 cm at $t = 0$, 17.1 cm at $t = 0.958$ day, and 17.9 cm at $t = 3$ days, according to the simulation. The volume of the cavity increased by 17.5 pct from its initial value, when measured immediately after the voltage treatment. After 2.042 days from the end of the voltage treatment, the cavity enlarged by 25.4 pct. Since the ACM to anode crust transformation is irreversible, this material does not recover after such thermal treatment. The proportion of ACM in the anode cover, [ACM], decreased from 47.7 to 44.1 pct immediately after the voltage increase and it got to 42.3 pct at $t = 3$ days. Furthermore, the total size of the anode cover decreased since the cavity enlarged. The side ledge, considered as a reversible phase change material, decreased by 43.9 pct after 0.958 day of voltage increase. At $t = 3$ days, it enlarged by 17.7 pct from its initial volume, which can be related to the thinner and more conductive anode cover.

Table IV. Effects of the Voltage Increase on the Side Ledge, the Cavity, and the ACM

	t (day)		
Simulation	0.000	0.958	3.000
Side ledge at BMI (pct)*	ref.	− 60.1	36.5
Volume of side ledge (pct)*	ref.	− 43.9	17.7
Height of cavity_{SC} (cm)	13.7	17.1	17.9
Volume of cavity in AEC (m^3)	0.610	0.717	0.765
Volume of cavity in AEC (pct)*	ref.	17.5	25.4
[ACM] in the anode cover (pct)	47.7	44.1	42.3

*Comparison with the initial condition ($t = 0$).

3. *Comparison with the measurements on the industrial AEC*

The instrumented anode A was measured during the voltage treatment which occurred at approximately 75 pct of its life. However, the anodes of the slice of AEC are at mid-life to reproduce the average behavior of an industrial cell. Hence, a difference is expected between predictions and measurements due to the different anode sizes (this effect is discussed in the Section III–B). The measured data taken in the anode crust at 0.0 cm ($T_{meas.,crust,0.0}$) and at 5.0 cm ($T_{meas.,crust,5.0}$), and the data taken on the anode stub next to the deck ($T_{meas.,stub,0.0}$ and $T_{meas.,stub,top}$) are shown in Figure 10(a). When compared with the predicted T_{model}, the measured temperatures in the anode crust showed a larger difference than the measurements on the stub. All $T_{meas.,crust}$ and $T_{meas.,stub}$ increased when the voltage increase started (at $t = 0$) and they decreased after this treatment. The temperatures on the metallic surfaces recovered their original values after the voltage treatment, while the temperatures inside the anode crust remained higher according to the measurements. For example, $T_{meas.,crust,0.0}$ started at 797 °C and $T_{model,crust,0.0}$ at 798 °C ($t = 0$), while they reached 881 °C and 812 °C, respectively, at the end of the voltage treatment. In both cases, the temperatures increased. However, in industrial cells the higher temperature of the liquid bath can also enhance the rate of bath vapor production and increase the gas temperature, which may have amplified the temperature inside the industrial anode crust.

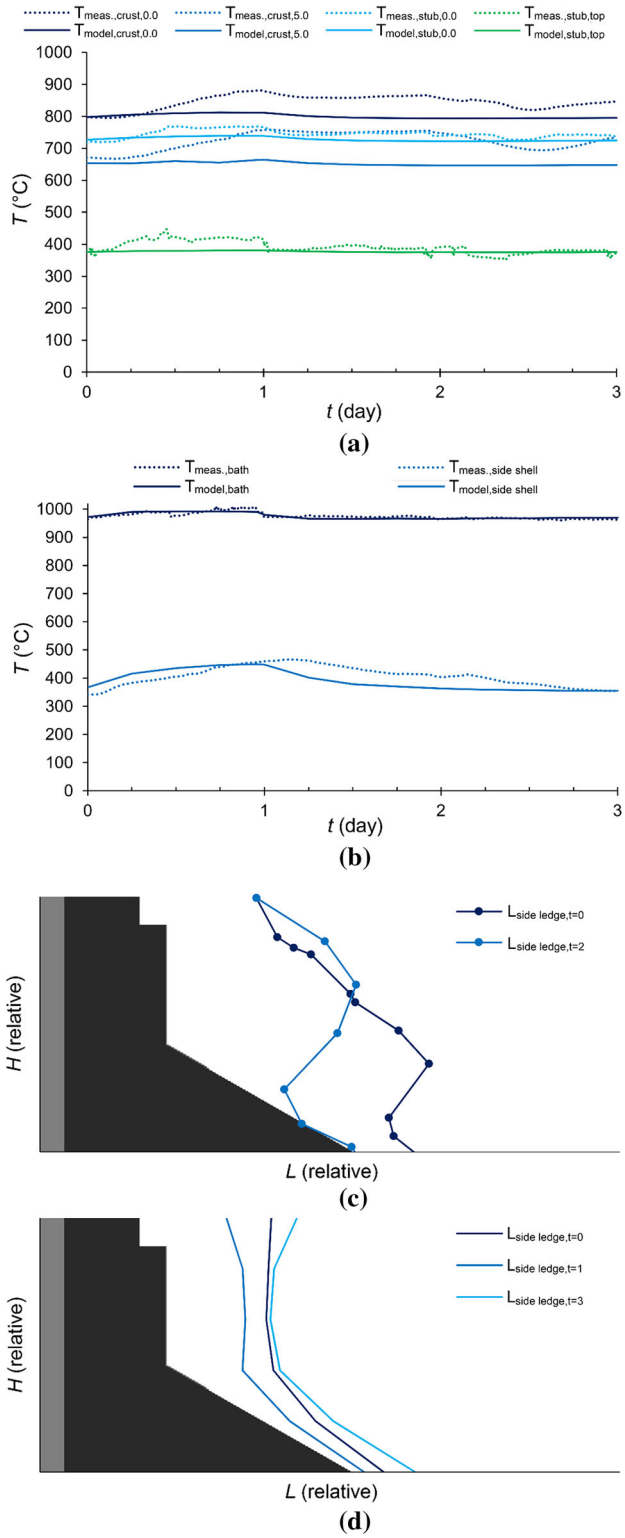


Fig. 10—(*a*) Temperatures taken in the anode crust and on the stub next to the deck. (*b*) Temperatures taken in the liquid bath and on the side shell at BMI. (*c*) Side ledge profile measured in an industrial AEC at $t = 0$ and $t = 2$ days. (*d*) Side ledge profile predicted with the slice of AEC at $t = 0$, $t = 1$ day and $t = 3$ days.

The liquid bath ($T_{\text{meas.,bath}}$) and side shell ($T_{\text{meas.,side shell}}$) temperatures were recorded by two thermocouples in order to evaluate the effect of the voltage increase, see Figure 10(b). The voltage increase (+ 8.5 pct of the AEC operating voltage) increased the temperature of the liquid bath in the industrial AEC by 29 °C from $t = 0$ to $t = 0.958$ day. In the numerical model, the temperature increased from 972 °C to 990 °C in the same time lapse. In both the measurement and the simulation, the temperature of the liquid bath recovered quickly after the voltage treatment. In one hour after the voltage increase, $T_{\text{meas.,bath}}$ decreased from 993 °C to 974 °C (− 19 °C), while $T_{\text{model,bath}}$ decreased from 990 °C to 980 °C (− 10 °C). After 7 h from the voltage treatment (at $t = 1.25$ day), $T_{\text{model,bath}}$ reached a slightly lower temperature (966 °C) compared to the initial state (972 °C). The $T_{\text{model,bath}}$ only regained its initial value after approximately 2 days from the voltage treatment. The $T_{\text{meas.,side shell}}$ and $T_{\text{model,side shell}}$ also increased due to the voltage increase and decreased slowly during the next 2 days. Accordingly, the $T_{\text{model,side shell}}$ were: 367 °C at $t = 0$, 449 °C at $t = 0.958$ day, 363 °C at $t = 2$ days, and 355 °C at $t = 3$ days. From $t = 2.75$ days, the $T_{\text{model,side shell}}$ was stabilized at a lower temperature (355 °C) than the initial value. The voltage increase affected the anode cover by increasing its thermal conductance due to thinner anode cover and lower concentration of ACM (see Table IV). With a more conductive cover, the top heat dissipation is increased and the side ledge adapts to this new thermal condition. Considering a similar external environment, the side shell temperature decreases due to this condition.

The side ledge thicknesses were evaluated in the industrial AEC and the transient model. Figure 10(c) shows the length (L) of the side ledge taken at various heights (H). $L_{\text{side ledge},t=0}$ is a typical side ledge profile which was determined from two measurements prior to the voltage increase, on anodes immediately next to the instrumented one. $L_{\text{side ledge},t=2}$ was measured at approximately 1 day after the voltage treatment on an anode next to the instrumented one to prevent disturbing the temperature recording. This measurement is presented since it is the closest to the voltage treatment. In the industrial AEC, the local side ledge thickness decreased due to the voltage increase which increased the temperature on the side shell at BMI. The length of the side ledge was also evaluated with the numerical model at $t = 0$, 1 and 3 days, as shown in Figure 10(d). The side ledge size decreased with increased voltage. Afterward, it enlarged to reach a larger size up upon the return to equilibrium, at $t = 3$. The temperature profile in the transient model and the side ledge evolution over time are presented in Supplementary Video S1 (see electronic supplementary material). In both the industrial measurements and the transient model, the side ledge decreased as expected with voltage increase. Moreover, a larger side ledge also implies a decrease of side heat losses due to its insulating behavior.

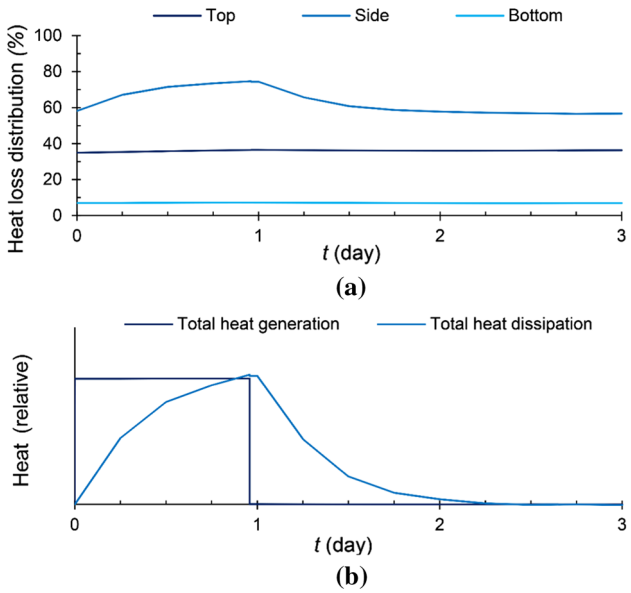


Fig. 11—(*a*) The heat loss distribution calculated in comparison with $t = 0$. (*b*) Total heat generation and heat dissipation in relative scale, also in comparison with $t = 0$.

4. *Impact of the voltage increase on the heat dissipation*

Immediately before increasing the voltage (at $t = 0$), the heat losses were distributed as follows: 34.9 pct from the top, 58.1 pct from the side, and 7.0 pct from the bottom. The top heat losses include the anode assembly and the anode cover, thus all the parts dissipating heat under the pot hoods. The deck dissipates heat toward the outside of the AEC, and this part is thus included in the side heat balance. Figure 11(a) demonstrates the effect of the voltage increase on the heat loss distribution when compared with the values taken at $t = 0$. After 0.958 day of increased voltage, the heat losses augmented to 36.5 pct (top), 74.4 pct (side), 7.2 pct (bottom), representing an increase of 18.1 pct from the initial heat losses. The side heat losses increased the most (+ 16.3 pct), while the top dissipation increased by 1.6 pct. Nevertheless, the top parts are mostly composed by irreversible phase change materials (ACM and anode crust), while the side ledge is a reversible phase change material. Accordingly, at $t = 3$ days, the top heat losses remained augmented at 36.3 pct, but the side heat dissipation was lower than the initial value, at 56.7 pct (− 1.4 pct). The bottom heat losses remained almost constant from $t = 0$ to $t = 3$ days.

Figure 11(b) shows the impact of the voltage increase on the heat generation (Joule heat) from the slice of AEC, and also the thermal lag between heat generation and dissipation in this transient model. The heat dissipation responds more slowly due to both latent heat in the phase change materials and low thermal conductivities. According to the transient model, it takes about 1 day for the AEC to stabilize almost entirely from this voltage increase, while the time required to reach the stabilization is close to 2 days. After the stabilization of the thermal dissipation, the AEC ends up in a new thermal balance implying higher heat losses from the top and lower from the side. This effect is amplified with increasing ACM conversion and anode crust melting in the anode cover.

B. *Anode Life Cycle Modeling*

1. *Temperature profile*

The thermal behavior of the top parts was investigated with a transient thermal-electric model which takes into account the carbon anode consumption and the anode assembly descent to maintain ACD during the electrolysis process. The anode life cycle is modeled by applying as nodal constraints the previously calculated liquid bath and side ledge temperatures, which were obtained at steady state with the slice of AEC. This second simulation depicts the typical evolution of an anode with its cover, thus without operational disruption such as voltage increase. The anode assembly and the anode cover located both OA and in the CC were moved downward during the anode life. The ACM started at T = 30 °C ($t = 0$), it transformed into anode crust according to Eq. [9] (between 685 °C to 710 °C), and the anode crust melted to form a cavity at 935 °C. Figure 12 shows the anode downward movement, the anode cover in the SC that remained fixed, and the temperature variation during the anode life (at $t = 0$, 25, 50, 75, and 100 pct of the total anode life), where $t = 100$ pct represents less than a month. Moreover, a video is provided in the electronic supplementary material (see Supplementary Video S2) to visualize the temperatures with smaller time differences and to see the mesh updating between each analysis step. It can be observed that the mesh adapts to the moving parts with the programmed moving geometry, mesh update, and nodal interpolation strategies. From $t = 75$ pct to $t = 100$ pct, the ACM_{OA} and $crust_{OA}$ were lowered below the $cavity_{SC}$, and accordingly an open cavity was generated using the radiosity module.

2. *Geometric evolution of the anode cover*

The ACM and anode crust profiles were validated in industrial cells with the measurement method previously shown in Figure 3(a), in order to built the right initial geometry and to confirm that both the anode covers OA and in the CC follow the anode displacements. Figure 13(a) shows the anode cover of the instrumented anode B with a picture taken immediately after the covering procedure. This instrumented anode operated typically, thus without any voltage treatment. In the simulations, the anode cover is designed based on the average thicknesses measured in industrial AEC. However, the leveling of the ACM in the industrial cells is not as uniform as in the model.

In this transient model, the body parts $crust_{SC}$ and the ACM_{OA} disconnected from each other at $t = 72$ pct of the total anode life. Accordingly, an open cavity was generated in the model, in contrast to the perfect cavity considered at $t < 72$ pct. In the industrial AEC, some flames are expected to be observed, as shown in

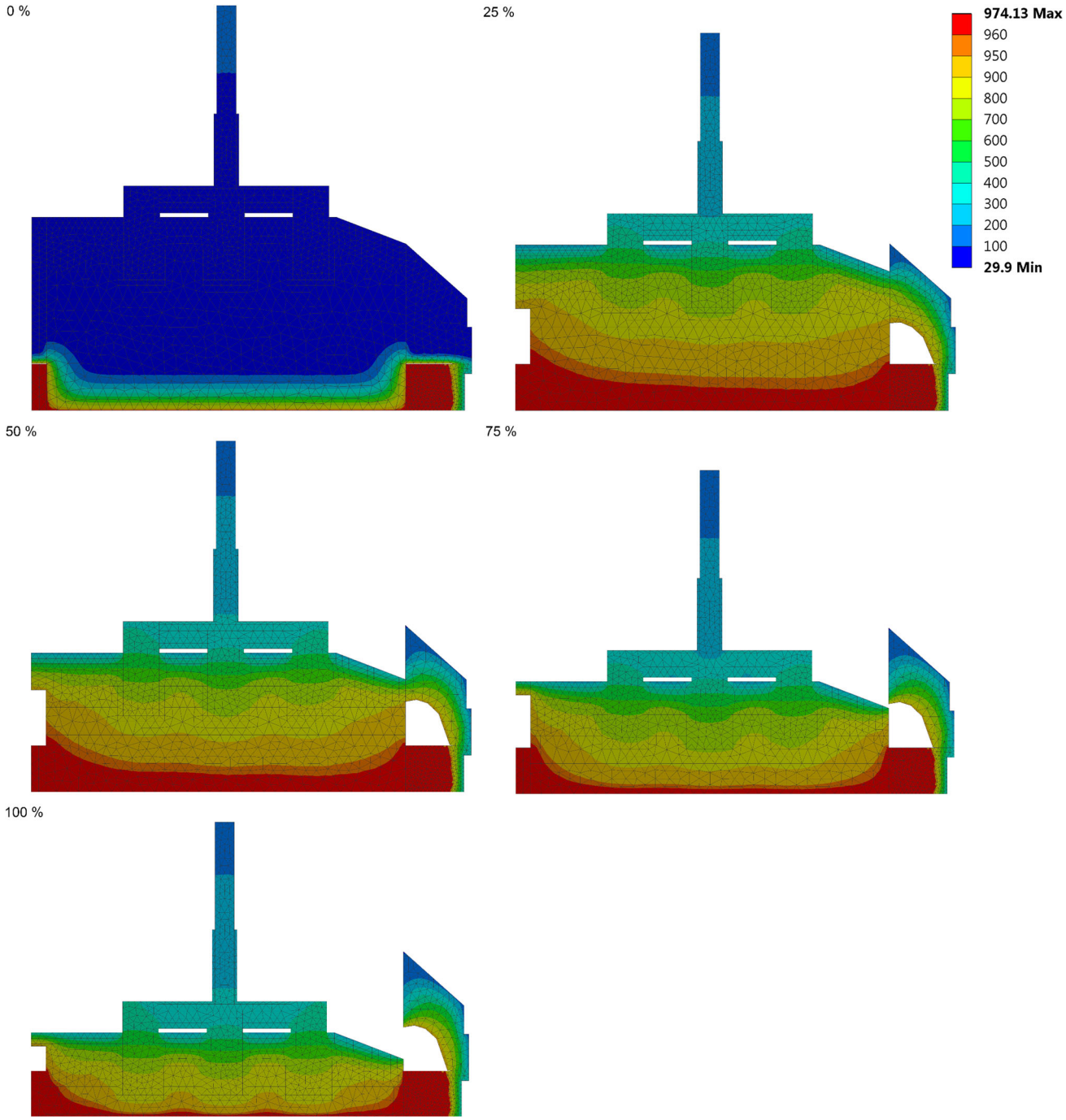


Fig. 12—The temperature profile (°C) at 0, 25, 50, 75, and 100 pct of the anode lifespan, predicted using the anode life cycle model.

Figure 13(b). The top boundary conditions were kept uniform during all of the anode life modeling, even though the flames are expected to increase the top T_{∞}.

3. *Comparison between the anode life cycle model and the instrumented anode B*

Different temperatures were measured with thermocouples during the anode life and these results are compared with the transient prediction. Figure 14 gives the temperatures recorded on the anode rod (a), on the closest stub to the deck (b), and in the ACM and anode crust (c). The average temperatures measured and predicted are described in Table V. The temperatures on the anode rod increased quickly between $t = 0$ to $t = 5$ pct and they reached 75 pct of the average test measurements after 5 pct of the anode life. The temperatures on the anode rod reached the average of the test after $t = 24$ pct. The temperature on the top of the steel stub ($T_{\text{meas.,stub,top}}$) followed the exact same trend than the anode rod: it reached 75 pct of the average value also at $t = 5$ pct and it reached its average at $t = 24$ pct. On a longer time scale, $T_{\text{meas.,stub,top}}$



(a) (b)

Fig. 13—(*a*) Instrumented anode immediately after the covering. (*b*) Another anode at $t = 80$ pct with a typical cover exhibiting flames.

increased slowly between $t = 25$ pct ($T = 375.5$ °C) and $t = 75$ pct ($T = 399.9$ °C). Therefore, the temperatures of the metallic parts of the anode assembly in contact with gas located under the pot hoods increased rapidly and they remained relatively stable around the mid-life of the anode. At $t = 77$ pct, the temperatures of these metallic parts began to raise abruptly.

The temperatures measured in the ACM and anode crust got to 75 pct of their average values at $t = 4$ pct for $T_{\text{meas.,crust,0.0}}$, at $t = 7$ pct for $T_{\text{meas.,crust,5.0}}$ and at $t = 10$ pct for $T_{\text{meas.,ACM,10.0}}$. In order to reach their averages, $T_{\text{meas.,crust,0.0}}$, $T_{\text{meas.,crust,5.0}}$, and $T_{\text{meas.,ACM,10.0}}$ took, respectively, 19, 43, and 67 pct of the anode life. Accordingly, the temperature rise in the anode cover was slower than the metallic parts, as expected due to their lower thermal conductivity and because of the endothermal transformation of ACM into crust. Moreover, the temperatures of the anode cover raised drastically at the end of the test (also at $t = 77$ pct) and it contributed to increase the average temperature. This rapid temperature rise may be explained by the smaller distance between the bottom of the anode crust and the top of the bulk liquid bath in the AEC. Consequently, high temperature liquid bath may enter in contact with the anode crust and this will increase the local temperature. However, the fluid dynamic and the related heat transport are not considered in the current thermal-electric model.

In a general perspective, the anode cycle model predicted the temperatures of the anode rod, steel stub, and anode cover with a mean absolute relative error (MARE) of 5.2 pct, considering the absolute temperature scale (Kelvin) for the error calculations. The MARE was calculated with the average temperatures in Table V (subtracting the measured value from the predicted, dividing by the measured, taking the absolute, and then calculating the mean of all the calculated absolute values). Moreover, the prediction accuracy is better at $t < 77$ pct (calculated MARE of 2.7 pct from all the dataset) and it decreases at the end of the anode life. The transient model predicted with a very high accuracy the temperature rise in the anode cover, especially from $t = 0$ to $t = 25$ pct of the anode life with a calculated MARE of 1.2 pct.

4. *Impact of the anode consumption on the cavity size and on the top dissipation*

The anode life cycle model allows to follow the evolution of the cavity and the top heat losses, which include the anode cover and the anode assembly. The height of the $\text{cavity}_{\text{SC}}$ was measured in industrial AEC at the location in Figure 3(b). This industrial measurement was subject to large variations (standard deviation of ± 3.2 cm), since the measurements were taken on other anodes to prevent the perturbation of the instrumented anode B. According to Figure 15(a), the height of the $\text{cavity}_{\text{SC}}$ increased up to $t = 38$ pct of the anode life and then remained stable until the end, which is due to the static behavior of the crust_{SC}. In the case of the $\text{cavity}_{\text{CC}}$, it increased up to $t = 15$ pct, stabilized until $t = 72$ pct, and then it decreased continuously until the end of anode life time. The height of the cavity in the CC and between the anodes decreased due to the downward movement of the anode assembly. In parallel, the crust was melting as it was closer to the bulk liquid bath.

The top heat dissipation, from the parts located under the pot hoods, was compared with the average prediction during all the anode life. Initially, the ACM and anode assembly were cooler than the pot gas, and thus these parts were heated. The top heat dissipation reached its average value at $t = 21$ pct, and then it oscillated due to the varying electric current of the anodes and increased up to the end of the anode life. The top heat dissipation at mid-life corresponded to 103 pct of the average, thus 3 pct higher. Accordingly, the investigation of mid-life anodes gives a proper approximation of the average top heat dissipation from all the anode life. Nevertheless, close to the end of the anode life, the transient model underestimated the top temperatures (as discussed in Section III–B–3), and hence the top heat dissipation in industrial AEC is expected to be higher.

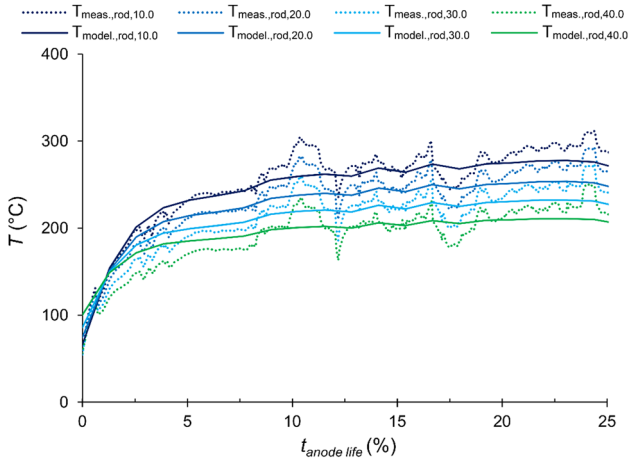


Fig. 14—(*a*) Temperatures on the anode rod taken at 10.0, 20.0, 30.0, and 40.0 cm from the yoke. (*b*) Temperatures measured at the bottom of the stub (0.0 cm) and on its top surface. (*c*) Temperatures recorded in the ACM and anode crust at 0.0, 5.0, and 10.0 cm from the top surface of the carbon anode. At the end of the test, only $T_{\text{meas.,ACM,10.0}}$ remained lower than the transformation temperature of ACM to anode crust.

C. *Significance of the Results for the Aluminum Industry*

1. *Voltage increases in industrial cells*

The effect of a substantial voltage increase (+ 8.5 pct over a long period of 23 h) was quantified by both the transient slice model and the industrial measurements. The voltage treatment caused the melting of 0.107 m^3 of anode crust, and thus 267.5 kg of material with high [Al_2O_3] has fallen into the bulk liquid bath, according to the prediction. This additional feeding of Al_2O_3 in the AEC might result in sludge formation at the cathode surface, if the liquid bath does not dissolve the melted anode crust.

The voltage increase also reduced the side ledge size, and consequently the side heat dissipation increased by 16.3 pct from its initial value. Additionally, this treatment deteriorated the anode cover and its thermal conductance increased. Since the ACM → anode crust → cavity transformations are irreversible, the top heat dissipation remained higher even once the operating voltage was brought back to its target value. Accordingly, the side ledge, which is reversibly produced from bath phase change, adapted to the increased top heat dissipation and reached a larger size than initially.

In industrial AEC, the combination of additional Al_2O_3 on the cathode surface and the additional heat losses from the side could result in Al_2O_3 precipitation on the bottom surface of the AEC. Moreover, the voltage increase augmented the liquid bath temperature, thus favoring vapor production ($NaAlF_4$). This evaporation increases the cryolite ratio and the bulk bath liquidus. Therefore, the side ledge and ledge toe can reach a larger size than predicted due to the increased liquidus temperature of the bath under real conditions.

2. *Anode life cycle and industrial events*

The complete life of the carbon anode with its metallic parts and covering materials has been simulated for the anode lifespan. This new transient thermal-electric model predicted accurately, with a MARE of 5.2 pct, the thermal behavior of the top parts. After 77 pct of the anode life, the modeling could be improved by adding mass transfer phenomena such as liquid and gas motions into the anode cover. Still, this transient model has predicted accurately the temperatures in the top parts for a time scale representing more than two weeks. The anode crust properties (k, ρ, c_p) have been determined recently up to 950 °C and it helped to obtain the proper temperature profile in this material.[9] Furthermore, the transient modeling allowed to predict both the ACM to anode crust conversion and the cavity growth in function of time, two phenomena that must be considered to accurately predict the top heat dissipation of AEC.

Operating events such as voltage treatments and anode covering practices can certainly change the thermal constraints on the anode cover, thus resulting in higher ACM → anode crust → cavity transformations with additional anode crust melting and falling toward the bulk liquid bath. During the anode life, the anode crust melts and feeds the AEC with uncontrolled

Table V. Average Temperatures Measured on the Instrumented Anode and Predicted with the Transient Model

Location	Average Temperature (°C)		
	Measurement	Simulation	Error (Pct)*
Rod at 10.0 cm	301.9	263.5	− 6.7
Rod at 20.0 cm	279.5	240.9	− 7.0
Rod at 30.0 cm	256.3	221.3	− 6.6
Rod at 40.0 cm	231.7	202.2	− 5.8
Stub at 0.0 cm	680.6	697.5	1.8
Stub top	391.5	364.1	− 4.1
Anode crust at 0.0 cm	768.3	776.4	0.8
Anode crust at 5.0 cm	630.0	601.8	− 3.1
ACM at 10.0 cm	429.7	351.8	− 11.1

*Error calculated using the Kelvin scale.

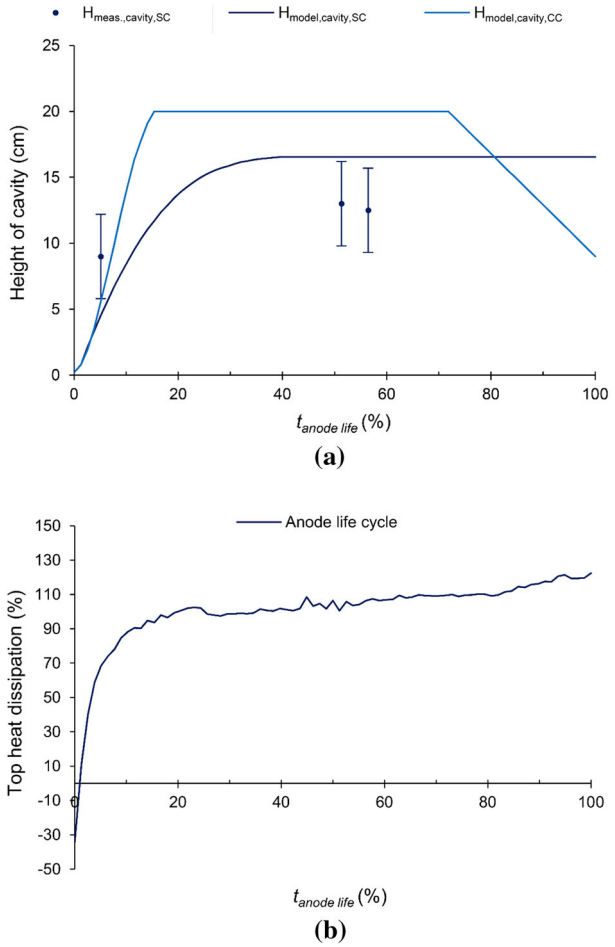


Fig. 15—(*a*) Height of the cavity during the anode life. (*b*) Top heat dissipation in percentage of the average value over the life time.

Al_2O_3 quantity. Accordingly, each additional thermal constraint on the anode cover must be considered for improving the industrial AEC control and operation. In summary, the heat generation and consequent heat dissipation have to be maintained stable to reduce anode cover deterioration.

D. *General Scheme of the Modeling Algorithm*

The approach for moving geometry, mesh update, and nodal interpolation was integrated in two different transient models in order to predict the thermal behavior of the AEC and the anode consumption throughout its lifespan. This modeling approach can be adapted to other finite element models that have to consider time-dependent domains, from the motion of solid bodies or moving boundaries, for example. Figure 16 illustrates the method with the meshes taken at two subsequent analysis steps. The downward movement due to anode consumption is remeshed for each analysis step; thereafter the previous nodal temperatures are interpolated and used as initial condition for the current step. The moving front of the anode crust melting is updated at each step and the radiation boundary adapts to the moved location in order to perform the radiosity calculations of each surface. With this method, the nodes remain connected for each analysis step and the mesh conforms to the quality criteria.

The approach for moving geometry, mesh update, and nodal interpolation can be applied to larger scale simulations such as the complete electrolysis cell with all its components. This complex modeling case would require important computer resources, but it would allow studying parameters like the complete anode change procedure and the positioning pattern. Moreover, it should permit to get closer to the actual behavior of the industrial AEC, in which the upper parts are moving. The modeling approach is advantageous when applied to simulation cases exhibiting significant deformations or displacements.

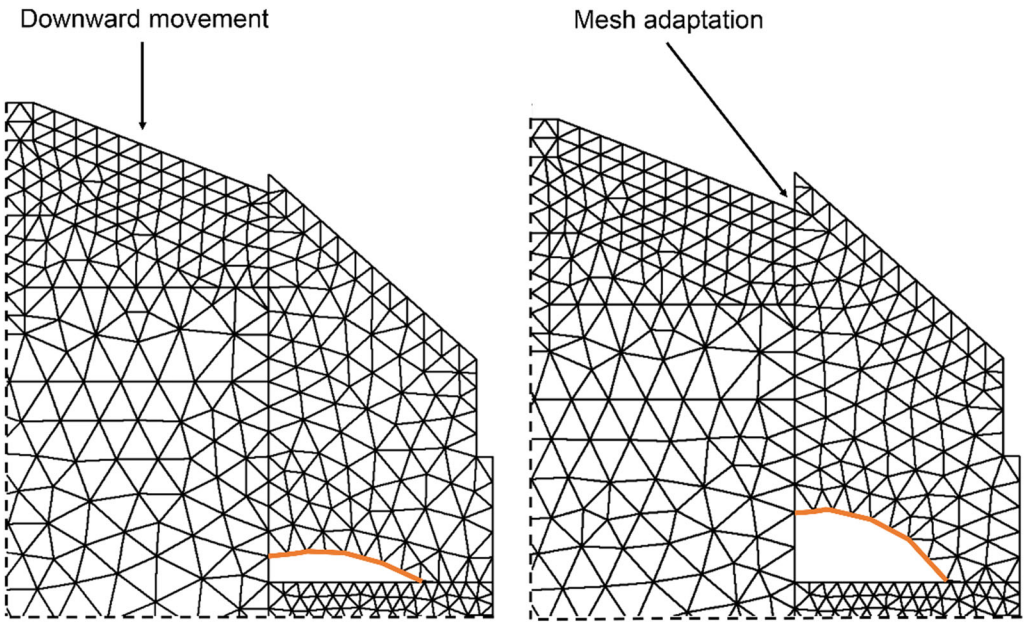


Fig. 16—General scheme of the moving geometry, mesh update, and nodal interpolation approach. The displacement of the moving boundary in orange.

IV. CONCLUSIONS

A new modeling approach was presented and applied to the transient thermal-electric modeling of a slice of AEC and the anode life cycle. Increasing the operating voltage by 8.5 pct for roughly one day affected the thermal balance of the AEC by melting the side ledge and deteriorating the anode cover, as confirmed by both modeling and industrial measurements. The top heat dissipation remained higher due to the irreversible transformations in the anode cover, even once the operating voltage was back at its target. In this occurrence, a new thermal balance has been reached and it was characterized by higher top heat losses (+ 1.4 pct of the total dissipation) and lower side heat losses. Accordingly, the anode crust melted to form a larger cavity and the side ledge adapted to the higher top dissipation by getting thicker.

For the first time, the complete thermal-electric evolution of the anode during all its lifespan was modeled by applying strategies for moving geometry, mesh update, and nodal interpolation. The anode life cycle model predicted the temperatures with a MARE of 2.7 pct, when compared with the installed thermocouples on the instrumented anode (for $t < 77$ pct of the anode life). The transient modeling of the temperature rise in the anode cover was particularly accurate considering the MARE of 1.2 pct. Therefore, this novel modeling method combined with the recently improved transient thermal properties (k and c_p of anode crust newly determined from 100 °C to 950 °C) and also the measured boundary conditions allowed to improve the thermal prediction of the top parts of the AEC. The modeling approach allows to consider geometric displacement with high mesh quality and may be applied to other finite element models dealing with large displacements or moving boundaries.

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CONFLICT OF INTEREST

The authors declare that they have no competing interests.

APPENDIX A

A modeling method for the geometry and mesh update algorithm followed by the nodal interpolation is detailed in this appendix. This method is applied to the transient thermal-electric modeling of the industrial AEC in order to consider the melting of the anode crust by a moving boundary.

A.1 PYTHON SCRIPTING IN ANSYS WORKBENCH

The moving geometry and mesh update algorithms are automated with a Python script interpreted by the ANSYS Workbench software. A while loop is programmed to solve each analysis step of the transient model and the results are saved automatically. The principal steps of this method are described by the scheme of Figure A1. A previously calculated or a new crust–cavity profile (moving boundary) is input in the parameter set to modify the geometry of the model.

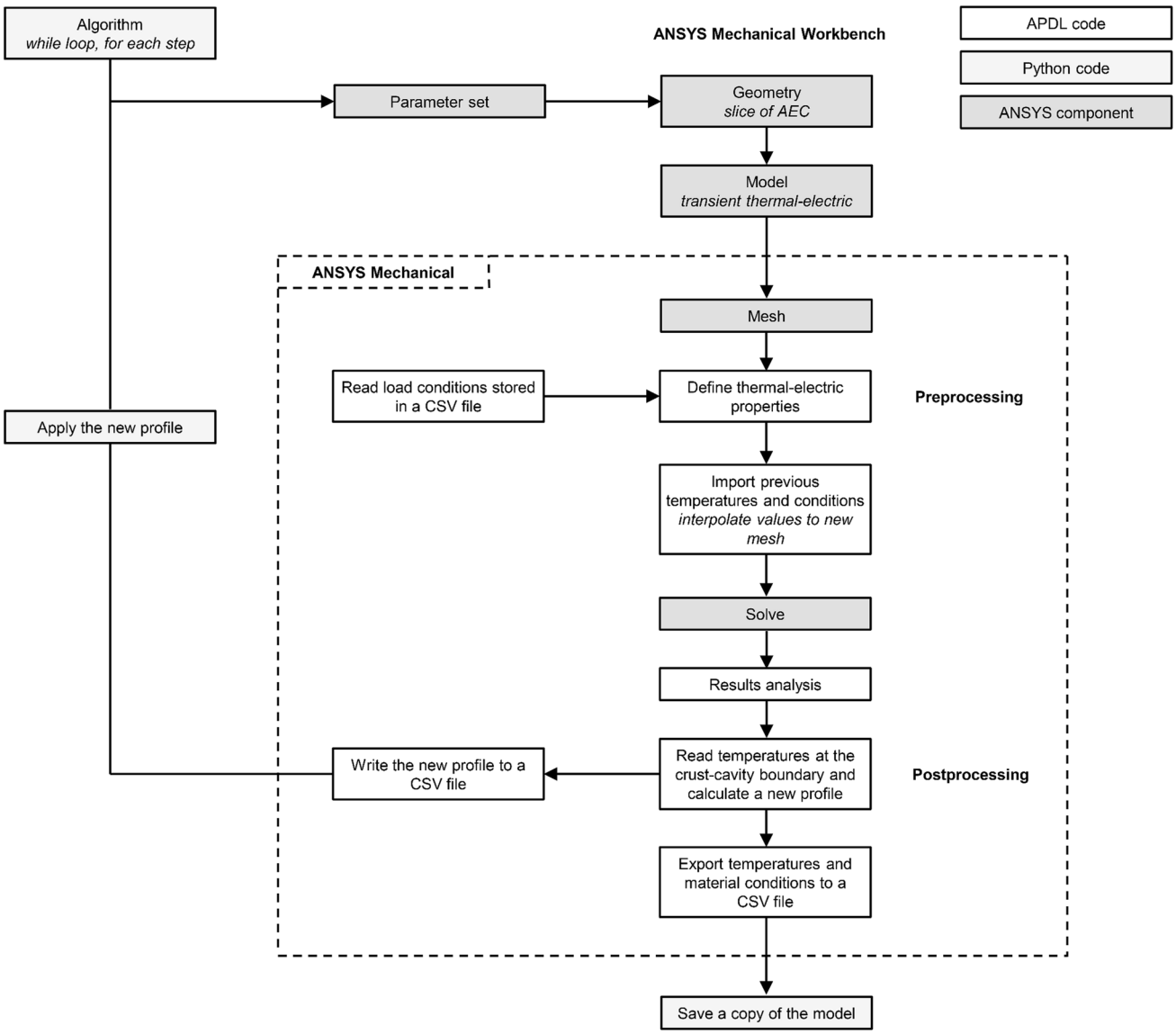


Fig. A1—Scheme of the modeling algorithm to solve the transient thermal-electric model with geometry and mesh updates, considering the moving boundary at the crust–cavity interface.

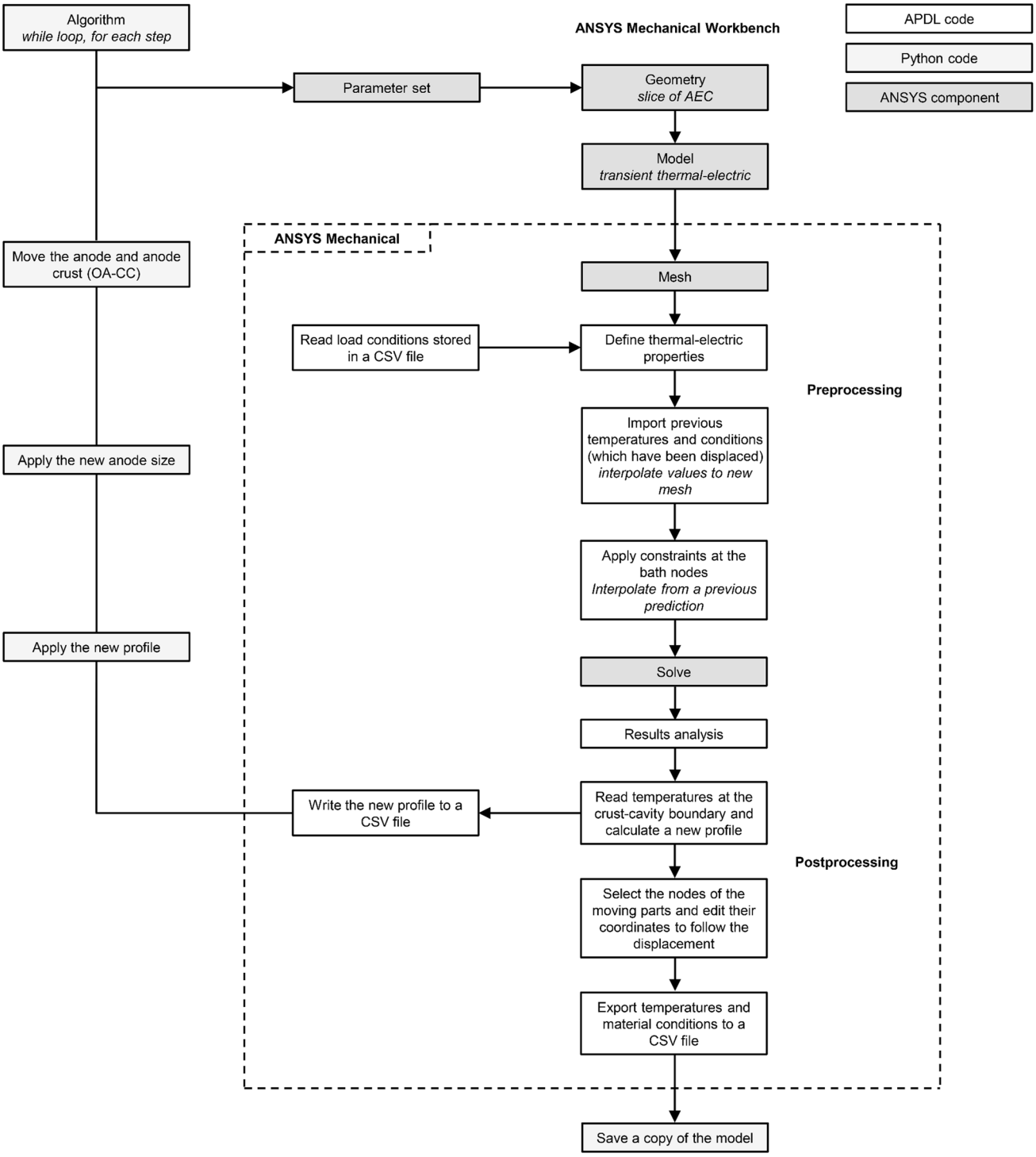


Fig. B1—Scheme of the modeling algorithm to solve the transient thermal-electric model with moving geometry, mesh update, and nodal interpolation, considering both the moving parts and the displacement of crust–cavity interface.

Accordingly, the geometry is updated using a Python command and a new mesh in the ANSYS Mechanical model is generated.

A.2 APDL SCRIPTING IN ANSYS MECHANICAL

The coupled thermal-electric analysis is performed using Mechanical APDL solver in the ANSYS Mechanical software. APDL commands are integrated to read the load data stored in a CSV file, like the thermal and electric properties or the boundary conditions. The new properties or conditions are applied by APDL commands to the current analysis. The previously calculated temperatures are mapped to the new mesh with an interpolation function (*MOPER).[43] This function performs matrix operations by comparing arrays, which contain the previous nodal coordinates and temperatures, and also the new mesh information. A 3D interpolation is applied to get the nodal temperatures on the new mesh. In that simulation case, a similar interpolation method is used to apply the previous material conditions to the newly generated elements. This step allows to consider the irreversible phase transformation in the anode cover (ACM $\rightarrow$ anode crust). With the previous temperatures and material conditions applied to the new mesh, the transient model is solved for a specified analysis time, before starting the postprocessing. Subsequently, the temperatures in the anode crust are read and a new crust–cavity profile is calculated in order to reach the melting isotherm. The new profile is written in a CSV file to be read by Python commands in order to update the geometric parameter and so on. The nodal coordinates and temperatures are also exported to a CSV file to be read in the next iteration. The procedure in Figure A1 is repeated up to the final time of the analysis.

APPENDIX B

A moving geometry, mesh update, and nodal interpolation method is developed to reflect the anode consumption and its downward displacement in the AEC. The modeling approach is automated with Python scripting and the results are interpolated from one step to the next one with APDL commands. A moving interface for the melting of the anode crust is integrated in this modeling method. This second case additionally integrates an algorithm to consider the time-dependent geometry related to the anode consumption. The motion of the nodes that are located inside the moving parts is governed by the velocity of anode assembly, a downward movement in this case. The node coordinates are modified to follow the anode displacement. Since the geometry is moved and the mesh is automatically updated, the new mesh conforms to the quality criterium initially imposed. The previously calculated nodal temperatures are then applied as initial conditions before the new analysis starts. The complete scheme of resolution is detailed in Figure B1.

B.1 PYTHON SCRIPTING IN ANSYS WORKBENCH

A new aspect of this second model is the displacement of the body parts with automatic updates of the CAD geometry by Python scripting. For each analysis time, a new position is calculated and applied to the moving parts. Since the anode is consumed by the process, its size is imposed before each resolution step. The python script also applies the new location of the moving boundary (anode crust melting) according to the post-processing results.

B.2 APDL SCRIPTING IN ANSYS MECHANICAL

In the ANSYS Mechanical interface, a new mesh is generated for each resolution step. The previously moved nodal and element properties are interpolated and then applied as initial conditions for the thermal-electric model. The boundary conditions and constraints for the analysis are imposed with APDL commands. Thereafter, the model is solved and the postprocessing is performed. During postprocessing, a new crust–cavity profile is calculated only if the anode crust has melted. The nodes in the moving geometry are selected and their coordinated are displaced using matrix operations to account for the downward movement. Consequently, the nodal temperatures and element properties are saved to CSV files to be imported in the next resolution step.

ELECTRONIC SUPPLEMENTARY MATERIAL

The online version of this article (https://doi.org/10.1007/s11663-019-01510-6) contains supplementary material, which is available to authorized users.

NOMENCLATURE

SYMBOL

A	Area (m^2)
c_p	Specific heat capacity (J/kg K)
E	Electric field (V/m)
F	View factor
H	Height (cm or m)

h	Convection heat transfer coefficient (W/m^2 K)
J	Current density (A/m^2)
k	Thermal conductivity (W/m K)
L	Length (cm or m)
q	Heat transfer rate (W)
q''	Heat flux (W/m^2)
$\dot{q}$	Rate of energy generation per unit volume (W/m^3)
t	Time (s, h or day)
T	Temperature (K or °C)
V	Electric potential or voltage (V)

GREEK

δ_{ij}	Kronecker delta
∇	Gradient vector field
ε	Emissivity
ρ	Density (kg/m^3)
σ	Electrical conductivity (S/m); Stefan–Boltzmann constant (W/m^2 K^4)

ABBREVIATION

ACD	Anode–cathode distance
ACM	Anode cover material
AEC	Aluminum electrolysis cell
BMI	Bath–metal interface
CAD	Computer-aided design
CC	Center channel
CFD	Computational fluid dynamics
CR	Cryolite ratio
HFS	Heat flux sensor
MARE	Mean absolute relative error
OA	On the anode
SC	Side channel

STYLE

bold	A vector
[brackets]	A matrix or a concentration of a chemical species

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